

# System Architecture & Multi-Agent Pipeline

End-to-end data flow from live spatial providers to queueing models and autonomous decision synthesis

## 1. INTERFACE & API GATEWAY LAYER

### Leaflet.js Spatial Map UI

Unified 2-in-1 Discovery & Plan SPA (/app)

### FastAPI Async Gateway

/api/search/\* & /api/advisor/\* REST Endpoints

### Google Cloud Run

Auto-scaling Serverless Container Runtime

## 2. AUTONOMOUS MULTI-AGENT ORCHESTRATION LAYER

### Orchestrator

- Task lifecycle
- Async routing
- Fallback coordinator

### Driver Agent

- Spatial discovery
- Connector filter
- Distance ranking

### Advisor Agent

- District extraction
- Competition matrix
- Zone synthesis

### Scoring Agent

- Multi-factor utility
- Grid & ROI analysis
- Viability ranking

### Data Agent

- BigQuery client
- Port occupancy
- Session telemetry

### Explainer Agent

- Gemini 2.0 Flash
- Natural reasoning
- Signal attribution

## 3. APPLIED QUEUEING & PREDICTIVE MACHINE LEARNING LAYER

### M/M/c Erlang C Queue Model

Stochastic p50 / p90 Wait-Time Distributions

### DemandForecaster (GBT Regressor)

Gradient Boosted Port Utilization Predictions

### QueueModelValidator

728-Config Invariant Assertion Mesh

## 4. LIVE DATA FUSION & HARDWARE PROTOCOL LAYER

### OpenChargeMap

- Global EV Registry
- Multi-station API
- Connector ratings

### OSM / Overpass

- Parking facilities
- Mobility hubs
- District polygons

### Google Places

- Commercial POIs
- Amenity density
- Spatial context

### NREL AFDC

- Federal station DB
- US-wide coverage
- Energy fallbacks

### Google BigQuery

- Session telemetry
- Port utilization
- Analytics dataset

### OCPP 1.6 (WS)

- Live port states
- Hardware socket
- Real-time sync